BioMS Zaku: algebraic decomposition and out-of-sample audit of indices and predictive equations

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Abstract. *Bioimpedance indices are usually validated by their correlation with a reference measurement, but almost all of them grow with body size and correlate with every tissue. BioMS Zaku is open-source software that audits these indices before they are accepted. It writes each index as a vector of exponents of the measured variables, predicts the correlation between two indices before computing either of them, from the covariance matrix of the logarithms, and tests out of sample whether the index predicts the target beyond what a declared negative control already predicts. On a full run of the package's example, the predicted correlation differed from the observed one by at most 0.015, a pair of published indices was identified as redundant ($\rho = 0.97$), and the audit separated indices specific to lean mass from indices that follow fat mass.*

1. Why I built it

I have always been drawn to bioimpedance, and the field has a standardization problem: methods, equipment and indices multiply, and they are compared in ways that are hard to repeat. I built BioMS Zaku to give the field a common, documented way to record indices and equations and to analyze them with rigor, so that modeling problems with bioimpedance data starts from a solid base. It is a public library, still in development.

2. The problem

An index that correlates with lean mass proves little: larger people have more of every tissue, and an index proportional to body size correlates with lean mass, fat mass and water at the same time. The bioimpedance literature keeps adding indices, many of them algebraic variations of others, validated by correlation with the reference. Three questions are missing before an index is accepted: is it new, or does it already exist under another name? Does it measure what it claims, or does it follow body size? Does it add anything to stature and body mass?

3. The method

Decomposition. Every index that is a product of powers of the measured variables (resistance R, reactance X_c, stature H and mass W) is represented by its vector of exponents a. If Σ is the covariance matrix of the logarithms of these variables in a population, the Pearson correlation between the logarithms of two indices is given, before either is computed, by

$$r(a, b) = \frac{a^\top \Sigma\, b}{\sqrt{a^\top \Sigma\, a \;\; b^\top \Sigma\, b}}, \tag{1}$$

an identity that holds for any distribution. Indices that are not exact products, such as the phase angle, receive a fitted vector, with the fit's R^2 declared. Observed redundancy is

measured with Spearman's correlation (threshold 0.95), and the earlier publication takes precedence.

Conditional negative control. The researcher declares a target (for example, the lean mass index from DXA) and a negative control (the fat mass index). On identical out-of-sample resamples, the software measures the gain of the index for predicting the target beyond the control (S1) and for predicting the control beyond the target (S2). The index is *specific* when there is S1 and no S2; it *tracks the control* in the opposite case; it *measures both* or *neither* in the others. A gain is present when its mean exceeds 0.03, its 95% interval excludes zero and the probability of it being positive is at least 0.95.

Added value and geometry. The gain over stature and body mass is also measured. Because target and control often come from the same scan and the same normalization, the software computes the cosine between their projections in the measured space and flags the contrast as weak by construction when it is high. Flags annotate the verdicts; they never change them.

Designing indices. A new index can be fitted on one partition and audited on another, never seen, like any published method.

4. Engineering and reproducibility

The Python package is released under the MIT license. A guided terminal interface leads the user without any programming, and a Jupyter notebook carries the example ready to run. The output is a self-contained HTML report: each result comes with how it was computed, how to read it and the rigor applied, and every table can be downloaded. Assumptions and thresholds are fixed in five normative contracts. The catalog holds 32 published methods, eight of which were checked against the primary source and are the only ones audited by default. Expressions are parsed by a closed grammar, without executing arbitrary code. A manifest records configuration, seeds, versions and the SHA-256 of every table, and two identical runs produce the same hashes. The suite has 246 test functions, with lessons calculated by hand, exact algebraic identities and synthetic cases with a known answer. The report comes out in Portuguese, English, Spanish or Italian, and its tables never contain individual-level data, so the analysis can run inside a partner's environment.

5. Demonstration on the package's example

The official example is a synthetic dataset of 1,400 people (700 per sex), generated from means and covariances estimated in NHANES 1999–2004 and reproducible byte for byte from the published parameters [CDC/NCHS 2004]. The run used the full preset (5×50 cross-validation and 2,000 resamples), with the lean mass index from DXA as the target and the fat mass index as the control, in each sex (strata 0 and 1). This report and its figures come from the public version of the code (commit 41fc62e).

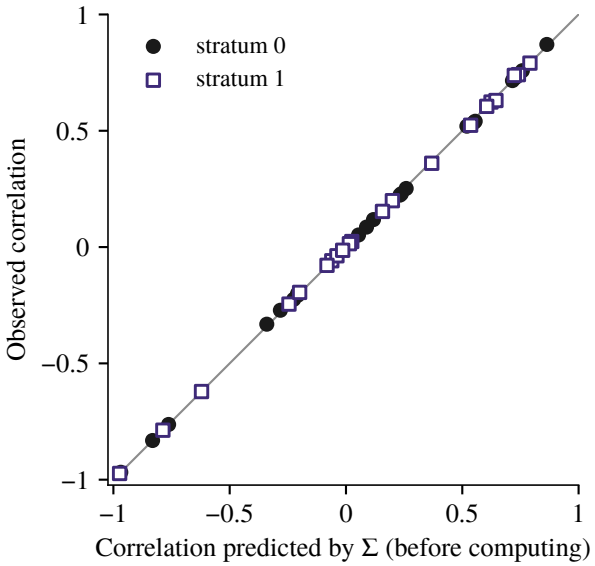


Figure 1. Correlation between the logarithms of pairs of indices: predicted by the matrix Σ, before computing the indices, against the observed one (42 pairs, two strata).

Predicted redundancy. The correlation predicted by Equation 1 differed from the observed one by at most 0.015 across the 42 pairs (Figure 1); for the indices that are exact products, the algebraic identity between the correlation and the geometry of the projections held to within 2×10^{-15}, at the limit of numerical precision. The component R/H [Piccoli et al. 1994] was identified as redundant with the impedance index H^2/R [Lukaski et al. 1985], published earlier ($\rho = 0.966$ and 0.968 in the two strata).

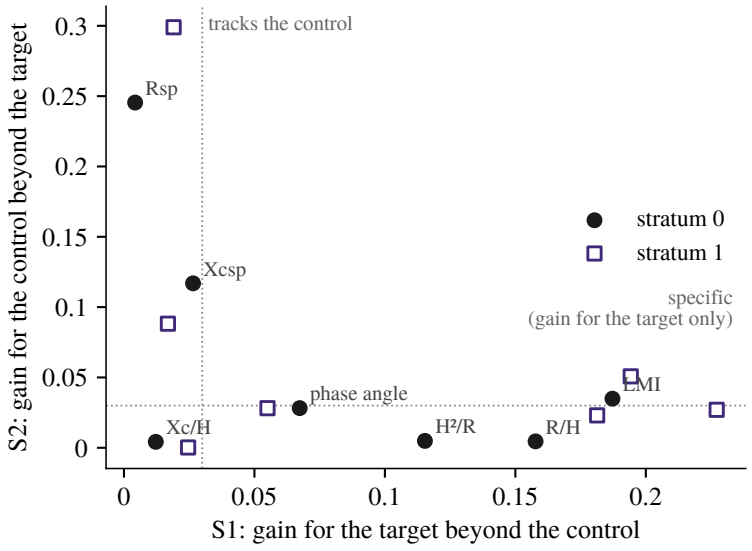


Figure 2. Audit against the negative control: gain for the target beyond the control (S1) and for the control beyond the target (S2). The dotted lines mark the 0.03 margin.

Audit. In each stratum, of the seven curated indices evaluated, three were specific

to lean mass, two tracked the control, one measured both and one showed no signal (Figure 2). Specific resistivity, for example, added almost nothing to the target (S1 of 0.004) and a lot to the control (S2 of 0.245) in stratum 0: it follows fat mass. The impedance index H^2/R was specific (S1 of 0.115; S2 of 0.005). Over stature and body mass, three indices in stratum 0 and four in stratum 1 added value.

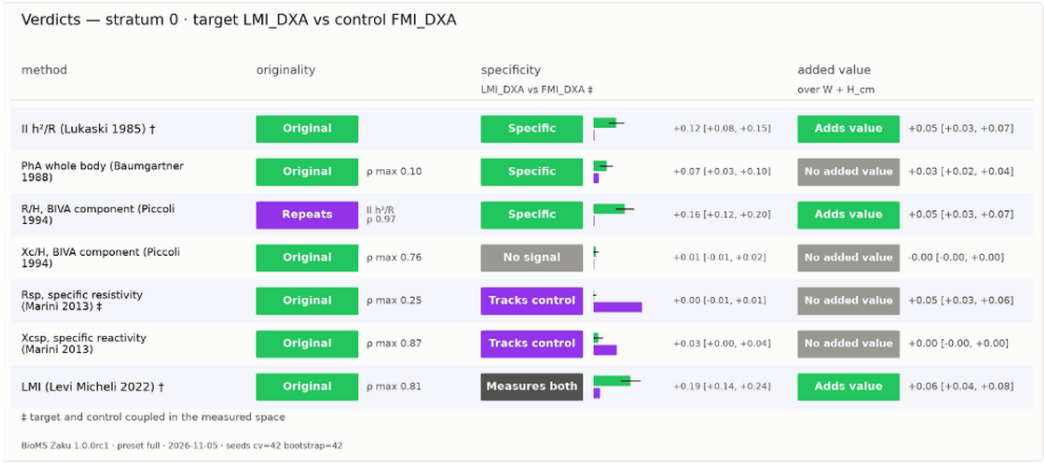


Figure 3. The verdict sheet as it appears in the HTML report the package generates (stratum 0): originality, specificity and added value for each method.

Honest verdicts. The software flagged the contrast in this example as weak by construction: target and control come from the same scan, and the cosine between their projections was 0.91 and 0.89. The sensitivity analysis showed that 5 of the 14 verdicts change somewhere on the grid of thresholds (margins from 0.02 to 0.05; probabilities from 0.90 to 0.99), and that 4 change when the logarithmic scale is replaced by the original one. These verdicts are borderline, and the report presents them that way.

6. Limitations

The demonstration uses synthetic data: the verdicts describe the example, not a real population, and ignore sampling weights by design. The software is at its first release candidate (1.0.0rc1) and has not been published to a package index. The algebraic decomposition is exact for indices that are products of powers; for the others, it depends on the quality of the fit, which is declared. High redundancy means that two indices carry the same information in that population, not that one outperforms the other.

7. Authorship and access

Lead creator and developer: T. P. L. Mota, with contributions from L. C. Oliveira Gonçalves and A. M. de Magalhães Neto. Open-source code: `https://github.com/bioms-zaku-framework/bioms-zaku`.

References

CDC/NCHS. National Health and Nutrition Examination Survey (NHANES) 1999–2004. Centers for Disease Control and Prevention.

Lukaski, H. C., Johnson, P. E., Bolonchuk, W. W. and Lykken, G. I. (1985). Assessment of fat-free mass using bioelectrical impedance measurements of the human body. *American Journal of Clinical Nutrition*, 41(4):810–817. doi:10.1093/ajcn/41.4.810.

Piccoli, A., Rossi, B., Pillon, L. and Bucciante, G. (1994). A new method for monitoring body fluid variation by bioimpedance analysis: the RXc graph. *Kidney International*, 46(2):534–539. doi:10.1038/ki.1994.305.